REHABILITATION & STRENGTH

Building Back Stronger

A 24-week sustainable strength program
for a recurrent L5/S1 disc herniation

Prepared for: Andreas
Date: 31 July 2026
Last revised: 17 October 2026
Training: 3 full-body gym sessions/week, ~60 min
Home: 15 minutes daily rehab + walking
Starting point: ~6 weeks post June 2026 flare, improving but symptomatic

> **Important.** This is an education and programming document, not medical advice. You have an active physiotherapist — bring this to them and let them veto, modify or re-sequence anything based on what they can see and feel that a document cannot. If you develop numbness in the saddle/groin area, changes in bladder or bowel control, or progressive weakness in one or both legs, stop training and seek emergency medical care the same day.

SECTION 1

The strategy, and why this one is different

You have had three episodes at the same level in under two years. That pattern is the single most important fact in this document, and it drives every decision in the program.

What recurrence usually means

Repeat herniations at one level are rarely caused by one bad rep. The common pattern is a mismatch between how fast *symptoms* settle and how fast *tissue tolerance* rebuilds. Pain typically resolves in weeks; the disc and the motor control around it take considerably longer. If you resume normal training and normal life loading at the point where you feel fine, you are loading a structure that is still behind. That is the trap, and it is where most people's second and third episodes come from.

The other common contributor is that rehab happens for 45 minutes in the gym and the other 23 hours are unchanged — sitting posture, how you pick things up, morning bending. Section 9 treats that as part of the program, not as an afterthought.

The five principles this program runs on

1 — Endurance before strength, strength before load

Trunk *endurance*, not maximal trunk strength, is what tracks with back resilience. The early phases build the ability to hold position for a long time under modest load. Heavy comes much later, and heavy is never the point.

2 — Load the hips hard, spare the spine

You can build serious muscle and functional strength with hip-dominant, torso-supported and single-leg work that puts very little shear or flexion demand on L5/S1. The program uses trap bar and elevated hinges, split squats, hip thrusts, chest-supported rows and loaded carries — all of which load the target tissue heavily while keeping your lumbar spine in neutral. Every squat, thrust and calf exercise is single-leg, so neither side can hide behind the other. The hinge stays two-legged on purpose: a symmetrical stance is what keeps the lumbar spine neutral under heavy load.

3 — Centralisation is the signal, not pain intensity

Symptoms moving *toward* your back and away from your leg means you are on the right track, even if back pain is unchanged. Symptoms moving *down the leg* means stop, regardless of how mild. This rule overrides how you feel about a given exercise.

4 — Boring, slow, and long is the fast route

The program is 24 weeks and deliberately under-ambitious in the first eight. Every previous episode bought you months. A conservative ramp that you complete beats an aggressive ramp that ends in a fourth herniation.

5 — Rebuild the nervous system, not just the muscle

An S1 nerve root that has been compressed leaves measurable deficits — calf strength, balance, and proprioception — plus a protective guarding pattern that outlasts the injury. Section 7 addresses this directly. This is the "back to 100% me" part.

What you are not doing

- **No loaded lumbar flexion.** No barbell from the floor with a rounding back, no sit-ups, no weighted crunches, no Roman chair flexion, no loaded toe-touches.

- **No loaded rotation** for the first 12 weeks. Anti-rotation only (Pallof press, carries).
- **No barbell back squat in Phases 1–2.** Axial compression through the spine plus the depth demand is the single worst risk/reward exercise for your situation. Split squats and step-ups give you the same quads with a fraction of the spinal cost.
- **No training to failure at any point in this program.** Form degrades near failure, and the form that degrades first is your lumbar spine.
- **No heavy bending or lifting in the first 60–90 minutes after waking.** Discs are at their most hydrated and least tolerant of flexion first thing in the morning.

SECTION 2

Rules of engagement

Read this before the first session. These rules decide what you do on any given day — they outrank whatever the program page says.

The traffic light system

Log the number, not the colour. You record your worst back pain since the last entry on a 0–10 scale — the same scale your physio uses — plus whether anything is happening below the knee. The band below follows from those two answers, and it is the band that drives the flare protocol and the phase-progression rules.

0 nothing at all · **3** noticeable, settles within a day · **6** limits what you do · **10** the worst you have had

GREEN 0-3/10	**No leg symptoms. Back stiffness or ache at or below 3/10 that settles within 24 hours.** Train as written. Progress as written. This is where the programme is designed to run.
AMBER 4-6/10	**Back pain 4-6/10, or stiffness that lingers more than 24 hours, but no symptoms below the knee.** Train, but: hold your loads where they are (do not add weight), cut the number of working sets by roughly a third, drop the hinge pattern for that session, and keep everything at RPE 5. Do not skip the session - movement helps here, load progression does not. Return to Green rules once you have had two consecutive Green days.
RED 7+/10, or any leg symptom	**Any pain, numbness, tingling or heaviness below the knee, whatever the number. Any new weakness. Symptoms spreading further down the leg than yesterday. Or back pain at 7/10 or above.** Stop loading. Walking, the daily home routine at reduced range, and nerve slides only. Contact your physio. The app opens a flare for you and rebuilds your sessions from stage 0 upward - see Section 8.

Symptoms moving toward your back are good news even if the number is unchanged. Symptoms moving down the leg mean stop, however mild and whatever the number says.

Red flags — same-day medical attention, not physio

- Numbness or altered sensation in the groin, genitals, or the area you would sit on a saddle
- New difficulty starting or stopping urination, or loss of bladder/bowel control
- Weakness in both legs, or progressive weakness in one
- Foot drop — catching your toes, unable to lift the front of your foot
- Sudden severe back pain with fever, or after a fall or trauma

Session rules

- **The rep that changes your form is the last rep.** If your lower back position shifts, the set is over — even if the number on the page says three more.
- **Brace before every rep.** Not a suck-in. A 360° stiffening — breathe into the belly and low ribs, then tension the whole abdominal wall as if bracing for a light punch. Hold it through the rep.
- **Warm up for five minutes, every session.** Non-negotiable: 3 min walk or bike, then 10 bodyweight hip hinges, 10 glute bridges, 8 cat-camel (gentle, no end range), 10 band pull-aparts.
- **RPE 5–6 means four or more reps left in the tank.** RPE 7–8 means two to three. You should never be closer than two reps to failure in this program.
- **Rest fully:** 2–3 min on main lifts, 60–90 s on accessories. Rushing rest is how form degrades.

- **If an exercise reproduces leg symptoms, remove it.** Not "push through it." Substitutions are in Section 6.
- **Single-leg work: weaker leg first.** The weaker leg sets the load and the reps; the stronger leg does the same, never more. Until the single-leg strength gap (Section 7) is under 10%, the weaker leg may do one extra set.
- **Supersets are marked SS1, SS2…** (Phase 3). Do one set of each, then rest 60–90 s. The pairs were chosen to fit the layout of your gym. If it is busy and you lose one station, do them as straight sets.

SECTION 3

Phase 1 — Re-entry (weeks 1–4)

Goal: establish a symptom-free baseline, rebuild trunk endurance and hip strength, and prove to your nervous system that loading is safe. You are not trying to get stronger in these four weeks. You are collecting evidence about what your back tolerates.

> **Rules for this phase.** Everything at RPE 5–6. No hip hinge under external load — the hinge pattern is trained through back extensions and pull-throughs, which give you the posterior chain without the lever arm. Controlled tempo: 2 seconds down, no bounce. If you finish a session thinking "that was easy," the phase is working as intended.

DAY A — single-leg squat emphasis · ~50 min

Exercise	Sets × Reps	RPE	Key coaching point
Step-up, low box (20–30cm)	3 × 10/leg	5–6	Drive through the top foot; do not push off the floor. Hold a rack for balance if you need it.
Machine chest press	3 × 12	6	Back flat against the pad throughout.
Chest-supported DB row	3 × 12	6	Incline bench takes your lower back out of the equation entirely.
Single-leg glute bridge	3 × 10/leg	6	Hug the other knee to your chest — that locks the pelvis and stops the low back arching. 2s squeeze at the top.
Single-leg calf raise	2 × 15/leg	7	S1 supplies your calf — this is rehab as well as training. Full range, 2s lower.
Half-kneeling Pallof press	3 × 8/side	6	3-second hold at full extension. Resist rotation; do not produce it.
Suitcase carry	3 × 30 s/side	6	Stay perfectly vertical. Weight is right when staying vertical is hard.

DAY B — posterior chain & upper body · ~45 min

Exercise	Sets × Reps	RPE	Key coaching point
Cable pull-through	3 × 12	6	Your hinge pattern practice. Short range, spine locked neutral.
Lat pulldown, neutral grip	3 × 12	6	Minimal lean-back. Keep the torso near vertical.
Incline dumbbell press	3 × 12	6	Feet flat, no arch. Set the bench at 30°.
Cable face pull	2 × 15	6	Upper back and rear delts; keeps posture honest.
Side plank (forearm)	3 × 20–30 s/side	6	Knees bent if the full version is hard. Completes the McGill "big three" with your home work.
45° back extension, bodyweight	3 × 12	6	Rise to **neutral only**, never past straight. Driven by glutes, not by the low back.

DAY C — single leg & upper body · ~50 min

Exercise	Sets × Reps	RPE	Key coaching point
Split squat (rear foot on floor)	3 × 10/leg	6	Hold a rack upright with one hand for balance. Torso upright.
Single-leg hip thrust	3 × 10/leg	6	Upper back on a bench, bodyweight to start. Chin tucked, ribs down — the movement happens at the hip, not the spine. Hips stay level; if one drops, the set is over.
Seated DB shoulder press	3 × 12	6	High back support, ribs down, no lumbar arch.
Seated cable row, neutral grip	3 × 12	6	Do not let your lower back round at the stretch. Shorten the range if it does.
Cable hip abduction	3 × 12/side	6	Glute medius. Weak abductors make the lumbar spine do pelvic stabilising work.
Cable lateral raise	3 × 15	6–7	Your only direct side-delt work.
Farmer carry (two hands)	3 × 40 s	6	Tall, braced, small steps.

Passing out of Phase 1
Move to Phase 2 when you have completed **four consecutive weeks with no Red days** and you finish sessions feeling the same or better than when you started. If week 4 has been anything other than clean, repeat weeks 3–4. Repeating a phase costs you two weeks; a fourth herniation costs you six months.

SECTION 4

Phase 2 — Build (weeks 5–12)

Goal: this is where the muscle gets built. Loads climb into a genuinely productive range, the hip hinge is reintroduced under load, and volume increases. RPE 6–8. Reps 6–12.

> **What changes.** The trap bar deadlift enters — from elevated blocks or the high handles, never from the floor in this phase. The elevated start position keeps your torso more upright and cuts the shear demand on your lumbar spine while still loading the hips and posterior chain hard. Dumbbell RDLs appear with a deliberately short range. Everything else gets heavier.

DAY A — hinge emphasis · ~60 min

Exercise	Sets × Reps	RPE	Key coaching point
Trap bar deadlift, from blocks	4 × 6	6–7	High handles or 10–15cm blocks. Brace, hips lead, bar path vertical. Reset every rep — no touch-and-go. **The one two-legged leg lift in the program, on purpose.** Film it from behind now and then: the hips must rise level, with no shift toward the stronger side.
Step-up, 40cm box	3 × 10/leg	7	Drive through the top foot; do not push off the floor. Highly functional.
Machine or DB flat press	3 × 10	7	Controlled, no bounce.
Chest-supported row	4 × 10	7	Heavier than Phase 1. Chest stays on the pad.
Standing single-leg leg curl	3 × 10/leg	7	Hamstrings at the knee. Cable with an ankle strap, or a standing leg-curl machine. The hip stays straight, which keeps the sciatic nerve slack — a seated curl bends the hip and stretches it. 2s lower.
Half-kneeling Pallof press	3 × 10/side	7	3s hold. Progress by moving further from the anchor before adding weight.
Suitcase carry	3 × 30 s/side	7	Add load steadily — this is one of the best trunk exercises you will do.

DAY B — upper body & glutes · ~60 min

Exercise	Sets × Reps	RPE	Key coaching point
Single-leg hip thrust	4 × 8/leg	7–8	Your main glute builder now. Upper back on a bench, dumbbell or plate held on the working hip. Hips level throughout.
Assisted pull-up or lat pulldown	4 × 8	7	Progress toward unassisted. Avoid kipping or swinging.
Incline dumbbell press	4 × 8	7	
Seated DB shoulder press	3 × 10	7	Back supported, ribs down.
Single-leg calf raise	3 × 12/leg	8	Hold a dumbbell on the working side once bodyweight is easy. Track it against the S1 benchmark in Section 7.

Exercise	Sets × Reps	RPE	Key coaching point
Side plank with top-leg lift	3 × 30 s/side	7	Progression from Phase 1's static hold.
Weighted 45° back extension	3 × 10	7	Plate on chest. Still to neutral only. Builds spinal-erector endurance, which is protective.

DAY C — single leg emphasis · ~60 min

Exercise	Sets × Reps	RPE	Key coaching point
Bulgarian split squat	4 × 8/leg	7	Dumbbells at your sides, not on your back. Brutal for the quads and glutes, near-zero spinal load.
B-stance dumbbell RDL	3 × 10/leg	6–7	Kickstand stance: back foot's toes on the floor for balance only, the front leg does the work. Short range — stop at mid-shin or wherever your back position starts to change. The bridge to Phase 3's single-leg RDL.
Dip (assisted) or flat DB press	3 × 10	7	
Single-arm DB or cable row	3 × 10/side	7	Supported on a bench. Also an anti-rotation demand.
Cable face pull	3 × 15	7	Rear delts and upper back; keeps posture honest.
Cable lateral raise	3 × 12–15	7–8	Your only direct side-delt work.
Farmer carry	3 × 40 s	7–8	Go heavy here. This builds grip, traps, and trunk simultaneously.
45° back extension hold	3 × 30–45 s	6–7	Bodyweight, held at **neutral** — a straight line from head to heels, never past it. Endurance, not strength: it is endurance of the back extensors that protects the spine. The same idea as your PT's static hold.

Passing out of Phase 2
Move to Phase 3 when you have completed **eight weeks with no more than two Amber days and zero Red days**, your trap bar has climbed meaningfully, and your single-leg calf raise benchmark (Section 7) is within 20% side to side.

SECTION 5

Phase 3 — Strength (weeks 13–24)

Goal: consolidate. Lower reps, heavier loads, more demanding variations — but the same spine-sparing exercise selection. This is the phase you can stay in indefinitely.

> **What changes.** Main lifts drop to 4–6 reps at RPE 7–8. Loaded rotation is introduced, carefully, through cable chops. The trap bar may come down toward the floor **only** if the previous eight weeks were entirely Green and you can maintain a neutral spine at that start position — check this with your physio, and film a set from the side. There is no requirement to ever pull from the floor.

DAY A — hinge emphasis · ~60 min

Exercise	Sets × Reps	RPE	Key coaching point
Trap bar deadlift	4 × 5	7–8	Lower the blocks only if the last 8 weeks were fully Green. Never grind a rep. **The one two-legged leg lift in the program, on purpose.** Film it from behind now and then: the hips must rise level, with no shift toward the stronger side.
Step-up, 40cm box	3 × 8/leg	8	Your Day A quad work, in place of the leg press. Heavier dumbbells than Phase 2; drive through the top foot.
Flat barbell or DB bench press	4 × 6	7–8	Modest arch only. Feet flat, glutes on the bench.
SS1 **Chest-supported row**	4 × 8	8	
SS1 **Single-leg calf raise**	3 × 12/leg	8	Symmetry is the target, not just strength.
SS2 **Standing single-leg leg curl**	3 × 8/leg	8	Hamstrings at the knee. Cable with an ankle strap, or a standing leg-curl machine. The hip stays straight, which keeps the sciatic nerve slack — a seated curl bends the hip and stretches it. 2s lower.
SS2 **Half-kneeling cable chop**	3 × 10/side	7	Rotation comes from the **thoracic spine and hips**. The lumbar spine stays quiet.
Heavy suitcase carry	3 × 30 s/side	8	

DAY B — upper body & glutes · ~60 min

Exercise	Sets × Reps	RPE	Key coaching point
SS1 **Single-leg hip thrust**	4 × 6/leg	8	Your heaviest glute work. Heavier dumbbell, or a Smith/machine single-leg setup. Hips level.
SS1 **Side plank with reach-through**	3 × 10/side	7	
Standing overhead press	4 × 6	7	Only if you can press without arching. Squeeze glutes, ribs down. Seated if not.
SS2 **Weighted pull-up or pulldown**	4 × 6	8	

Exercise	Sets × Reps	RPE	Key coaching point
SS2 **Kneeling ab rollout**	3 × 8	7	From the knees, short range. Roll out only as far as you can keep the ribs down and the hips in line. **Stop before the hips sag** — that is the low back dropping into loaded extension. Extend the range slowly over the weeks; it never has to reach full extension.
Incline DB press	3 × 10	8	
Weighted back extension	3 × 10	7–8	To neutral. Slow eccentric.

DAY C — single leg emphasis · ~55 min

Exercise	Sets × Reps	RPE	Key coaching point
SS1 **Bulgarian split squat**	4 × 6/leg	8	Heavier dumbbells, fewer reps.
SS1 **Dip or close-grip press**	3 × 8	8	
Single-leg RDL, dumbbell	3 × 8/leg	7	Hip hinge, balance and glute-medius work in one. Light load, high control.
SS2 **Single-arm cable row**	3 × 10/side	8	
SS2 **Cable lateral raise**	3 × 12–15	8	Your only direct side-delt work.
Cable face pull	3 × 15	7–8	
Heavy farmer carry	4 × 40 s	8	
45° back extension hold	3 × 45–60 s	7	Bodyweight, held at **neutral** — a straight line from head to heels, never past it. Endurance, not strength: it is endurance of the back extensors that protects the spine. The same idea as your PT's static hold.

SECTION 6

Progression, deloads and substitutions

How to add weight — double progression

Never add weight because a week has passed. Add weight when the current weight has been earned.

1. Start at the bottom of the prescribed rep range with a weight at the prescribed RPE.
2. Each week, add reps until you hit the top of the range on every set, at the same RPE.
3. Then add load — 2.5–5 kg on lower-body lifts, 1–2.5 kg on upper-body — and drop back to the bottom of the range.
4. Repeat. Expect this cycle to take 2–4 weeks per lift. That is normal and correct.

> **The one progression rule that matters most.** Only progress a lift if the previous week was Green. An Amber week freezes all loads. This single rule is the difference between this program and whatever you were doing before March 2026.

Deloads

Take a scheduled deload in **weeks 10, 18 and 24** — and additionally any time you have three or more Amber days in a fortnight. Phase 1 is light enough that it does not need one. On a deload week: keep the same exercises, cut working sets by half, and cap everything at RPE 5. Do not skip these. They are when tissue actually adapts, and they are cheap insurance against the accumulation pattern that produces recurrences.

Substitution table

If an exercise reproduces leg symptoms, swap it — do not push through, and do not just delete it.

If this hurts	Swap to	Why it works
Trap bar deadlift	Hip thrust + back extension	Same posterior chain, zero spinal shear from a loaded hinge.
Any hinge (RDL, pull-through)	Hip thrust, glute bridge, reverse hyper	Removes the flexion moment entirely while still loading glutes and hamstrings.
Leg press, seated leg curl	Step-up, split squat, standing leg curl	Both bend the hip with the pelvis fixed, which tensions the sciatic nerve. Upright single-leg work loads the same muscles with the nerve slack.
Seated cable row	Chest-supported row	Removes the need for your spine to resist the load at all.
Standing overhead press	Seated DB press, back supported	Eliminates the lumbar arch that standing pressing invites.
Bulgarian split squat	Rear-foot-down split squat or step-up	Less hip-flexor stretch, less pelvic tilt demand.
Side plank	Side plank from the knees, or standing Pallof	Same anti-lateral-flexion demand, lower total load.
Cable lateral raise	Seated DB lateral raise, or machine lateral raise	Seated removes any temptation to lean away from the load.
Ab rollout	Dead bug with a longer lever, or a front plank	Same anti-extension job with a fraction of the spinal load.

If this hurts	Swap to	Why it works
Everything (a Red week)	Walking, nerve slides, Big Three at reduced range	See Section 8.

SECTION 7

The daily 15 minutes at home

Do this every day, including gym days. Consistency here matters more than anything in the gym — it is what keeps the motor control you build from leaking away between sessions. If a day is chaotic, do the first four items and skip the rest.

#	Exercise	Dose	Notes
1	**Sciatic nerve slide**	2 × 10/ leg	Sit tall, slump slightly. Straighten the knee while you look up; bend the knee while you look down. Slow and smooth. This should feel like gentle traffic through the leg, never like a stretch or a symptom. Back off immediately if it does.
2	**Prone press-up**	10 reps	Lie face down, press your upper body up on your hands, hips stay down. Hold 2s. **Only if extension makes you feel better** — most L5/S1 disc presentations respond to extension, but test it. If it worsens symptoms, replace with 90/90 breathing.
3	**90/90 diaphragmatic breathing**	8 breaths	On your back, feet on a wall or chair, knees at 90°. Breathe into your belly and low ribs, long slow exhale. Resets a guarded, over-braced trunk.
4	**Glute bridge**	15 reps	Ribs down, squeeze at the top, 2s hold.
5	**Dead bug**	8/side	You already do this. Keep the low back flat against the floor — that is the whole exercise.
6	**Bird dog**	8/side, 5s hold	You already do this. Draw a small square with the extended hand and foot for a progression.
7	**Side plank**	2 × 25 s/ side	Completes the McGill “big three”. Knees bent version is fine.
8	**Half-kneeling hip flexor stretch**	2 × 30 s/ side	Tuck the pelvis first, then lean. Tight hip flexors make your lumbar spine compensate all day.

Walking — the highest-value thing you can do

20–30 minutes daily, at a brisk pace, arms swinging, separate from the routine above.

Walking loads and unloads the disc rhythmically, pumps fluid through it, and is the single most consistently useful intervention for disc-related back pain. Brisk matters — a slow shuffle does not have the same effect. On a Red day, walking is often the only thing you should do, and you should still do it.

Nervous system work — getting back to 100%

An L5/S1 herniation compressing the S1 nerve root leaves deficits that do not resolve just because the pain did. Three of them are worth targeting deliberately.

1. Calf strength — the S1 signature

S1 supplies your calf. The classic residual deficit after an S1 root problem is a weaker calf on the affected side, and most people never notice because two-legged calf raises hide it.

> **Benchmark: single-leg calf raise test**
>
> Stand on one leg, hand on a wall for balance only, and do full-range calf raises to failure. Test both sides. **Target: 25+ reps, and less than 20% difference between sides.** Test on day 1, then every four weeks. Record it. This is the most objective marker of nerve recovery you can measure yourself, and watching it improve is worth more for your confidence than any amount of reassurance.

2. Balance and proprioception

Nerve compression blunts the position sense coming from your foot and ankle, which your body compensates for by stiffening the trunk. Add to your daily routine, two minutes total:

Exercise	Dose	Notes
Single-leg stand	30 s per leg	Progress: eyes open on the floor → eyes closed on the floor → eyes open on a cushion → eyes closed on a cushion.
Heel walks and toe walks	20 m each	Simple, and directly tests the L5 and S1 myotomes.

3. Desensitisation and graded exposure

After three episodes, some degree of movement fear is entirely rational — and also now part of the problem. A back that is guarded all day is a stiff back that distributes load badly. The mechanism is well described: the more you avoid a movement, the more threatening your nervous system rates it, and the less you do it, the less tolerance you have for it.

The countermeasure is graded exposure, not avoidance and not bravado:

- Pick one movement you avoid — bending to tie shoes, lifting a bag from the floor, rotating to reach the back seat of the car.
- Do a small, controlled, unloaded version of it daily. Five reps. Nothing heroic.
- Each week, add slightly more range or slightly more load.
- Understand that some discomfort during graded exposure is expected and is not tissue damage. The rule that decides whether to continue is still centralisation vs. peripheralisation, not intensity.

Your spine is designed to bend. The long-term goal is not a back you protect forever — it is a back that tolerates ordinary life without you thinking about it. That takes deliberate, gradual re-exposure.

SECTION 8

The flare-up protocol

You will have a flare at some point in 24 weeks. Having a written plan for it is the difference between losing a week and losing a season — and it stops you making the panic decision to stop moving entirely, which reliably makes things worse.

Stage 0 — Settle

From the moment it starts. No gym. Walking, reduced-range home routine, nerve slides. *(no gym)*

- Do not go to bed. Bed rest beyond a day makes disc-related pain worse, not better.
- Walk short and often - 5-10 minutes every couple of hours beats one long walk.
- Find your position of relief and use it: prone, prone on elbows, or standing.
- Nerve slides gently, 2 x 10 per leg, twice a day. Reduce range until there is no provocation at all.
- Avoid prolonged sitting entirely. Stand or lie down.
- Contact your physio now. Do not wait to see whether it settles on its own.

Stage 1 — Reintroduce

After 3 consecutive clear days. Back in the gym, spine-neutral work only, half your pre-flare loads. *(50% of pre-flare loads · max 2 working sets · RPE 4-5 · Phase 1 exercises)*

- Supported and machine work only - nothing that loads the spine through range.
- Two working sets, never more. You should leave feeling you could have done twice as much.
- Home routine daily at full range again, walks toward 20-30 minutes.
- Any leg symptom during or within 24h of a session sends you back to Settle.

Stage 2 — Rebuild

After 7 consecutive clear days. Phase 1 as written minus the hinge, at 70% of pre-flare loads. *(70% of pre-flare loads · max 3 working sets · RPE 5-6 · Phase 1 exercises)*

- Squat pattern, carries and rows return. The loaded hinge does not.
- Run this for at least a week even if everything feels fine.
- Re-test the single-leg calf raise and compare it to your last benchmark.

Stage 3 — Re-load

After 14 consecutive clear days. Your normal phase, hinge still held back, 85% of pre-flare loads. *(85% of pre-flare loads · RPE 6-7 · your current phase)*

- Back to the phase you were in, with the heavy hinge and loaded rotation still parked.
- Sets as written. Loads are the brake here, not volume.
- Write the debrief now if you have not already.

Stage 4 — Full programme

After 21 consecutive clear days. Everything back, 95% of pre-flare loads. One clear week from closing the flare. *(95% of pre-flare loads · your current phase)*

- The hinge returns at the lightest load you would call productive, and climbs from there.
- Do not chase the number you were at before the flare. It will come back faster than you think.
- After a clear week here the flare is closed and you are back on the normal progression.

The phone app applies this ladder for you: log the flare and it rebuilds each session to the stage you are on, prefilling the reduced loads. A clear day is a day with no Red logged and no symptoms below the knee. Any

Red session, Red morning check-in, or session logged with leg symptoms resets the count to zero and drops you back to the stage that count belongs to.

> **The debrief — do this every time.** Within a week of any flare, write down: what you did in the 72 hours before it; whether the gym or daily life was the more likely cause; and what specifically you will change. Three episodes without a written debrief means the pattern that caused them is still invisible to you. This takes ten minutes and is probably the highest-leverage thing in this entire document.

SECTION 9

The other 23 hours

Your training is roughly 3 hours a week. Your life is 165. If the daily loading is not addressed, a good program cannot outrun it — and for recurrent herniations, daily life is more often the culprit than the gym.

Mornings

Your discs absorb fluid overnight and are at their most pressurised and least tolerant of bending for roughly the first hour after you get up. This is when a large share of disc injuries actually happen, doing something trivial.

- **No heavy bending or lifting for 60–90 minutes after waking.** Not the laundry basket, not the child, not the toolbox.
- Get out of bed by rolling to your side, then pushing up with your arms — do not sit straight up.
- Brush your teeth standing tall, or with one foot up on the toilet seat or a small step. Do not lean over the sink with a rounded back.
- **Do not train first thing in the morning.** Mid-morning or later is meaningfully safer for you.

Sitting

- **Break sitting every 30 minutes.** Stand up, walk 60 seconds. Set a timer if needed — this is the single highest-return habit change on this page.
- Sit fully back in the chair with support behind your lower back. A rolled towel works as well as an expensive cushion.
- Screen at eye height, feet flat on the floor.
- Long drives: stop every hour and walk. Use lumbar support — car seats are almost universally shaped to put you in flexion.

Lifting anything, ever

- **Hinge, don't round.** Hips back, chest forward, spine neutral. This applies to a 40 kg trap bar and to a dropped sock equally — arguably more to the sock, because that is when you are not paying attention.
- Get close to the object before you lift it. The further from your body, the greater the lever arm on your spine.
- **Never lift and twist in the same movement.** Turn with your feet.
- Split stance or half-kneel for low objects — one foot forward, or a knee down.

Sleep

- Side-lying with a pillow between your knees, or on your back with a pillow under your knees.
- Avoid sleeping face down if it stiffens you up, though for some people with an extension preference this is comfortable — go by how you feel in the morning.
- A very soft mattress that lets your hips sink allows sustained flexion for eight hours. Medium-firm is generally the better bet.

Body composition and general health

Two things reliably influence disc health and are worth naming: smoking, which measurably impairs disc nutrition and is associated with faster degeneration, and excess abdominal mass, which increases the sustained load on the lumbar spine. Neither may apply to you — but if either does, they are worth more attention than any exercise selection detail in this document. Adequate protein (roughly 1.6–2.2 g per kg bodyweight per day) and adequate sleep also directly affect how well you adapt to the training above.

SECTION 10

Tracking, milestones and next steps

What to record after every session

Two minutes in the tracker app. This is what turns 24 weeks of training into 24 weeks of information.

- Traffic light colour, before and the morning after
- Sets, reps and load for every exercise
- Any symptom below the knee — location, and whether it moved up or down the leg compared to last time
- Sleep hours and anything unusual in the previous 24 hours

Benchmarks — test on day 1, then every four weeks

Test	How	Target
Single-leg calf raise	Full range to failure, one leg, hand on wall for balance only	25+ reps, <20% side-to-side difference
Single-leg strength gap	Bulgarian split squat, same dumbbells on each leg (a load you could do ~12 with), weaker leg first; stop at the first rep where speed or form changes	<10% side-to-side difference
Side plank hold	Forearm side plank, both sides, to failure	60 s+, near-symmetrical
Bird dog hold	Longest clean hold, no hip drop	30 s+ per side
Single-leg balance, eyes closed	On the floor, hands on hips	20 s+, both sides
Walking tolerance	Longest continuous brisk walk without symptoms	45 min+
Sitting tolerance	Longest continuous sit before symptoms	60 min+

What "back to 100%" realistically looks like

By the end of week 24, a reasonable expectation is: no leg symptoms in normal life, symmetrical calf strength, the ability to pick things up off the floor without thinking about it, meaningfully more muscle than you have now, and a trap bar deadlift and hip thrust that are genuinely strong. What is *not* on the list is a one-rep-max back squat or a floor deadlift PR — and by design. Those specific lifts are where your risk lives, and they are not required for any of the outcomes you said you want.

Questions for your physio

Bring this document to your next appointment and specifically ask:

☐ Do I have a directional preference? Does extension centralise my symptoms, or does flexion?

☐ Is there any measurable S1 motor deficit right now — calf strength, ankle reflex, sensation on the outside of the foot?

☐ Given three episodes in two years, is a repeat scan or a surgical opinion warranted, or is continued conservative management still the right call?

☐ Is a loaded hip hinge appropriate for me at week 5, and do you want the trap bar higher than I have planned?

☐ Watch my hinge and squat and tell me what you see — can I have a video review at week 4?

☐ Are there specific movements you want me to avoid that are not on my list?

CLOSING

A note on how to hold all this

The temptation after three episodes is to train timidly forever, or to decide the back is broken and train around it. Neither works. Discs tolerate load — they need it. What they do not tolerate is load that arrives faster than adaptation, and load applied through a spine that is bent when it should be braced. This program is built around those two facts, and almost everything else in it follows from them.

Revision history

Date	Change
31 July 2026	Original 24-week program issued.
3 August 2026	Cable lateral raise added to Day A and Day C in all three phases. The program had no direct medial deltoid work; presses cover the anterior head and face pulls the posterior head. Placed on the two shortest/least shoulder-loaded days, and folded into the Day C cable superset as a tri-set, so total added session time is roughly five minutes a week.
6 August 2026	Cable lateral raise removed from Day A in all three phases; Day A was running well over its estimated time, and the raise was the only exercise there not serving the squat, hinge or trunk brief. It stays on Day C, where it is tri-set into the existing cable station and costs no extra time. Phase 1 Day C raised from 2 to 3 sets to partly offset the lost volume; Phases 2 and 3 stay at 3 sets. Side-delt frequency drops from twice to once a week, which is an acceptable trade for a single-joint isolation movement.
29 Aug 2026	Every exercise tagged with a muscle group and a spinal-load class. Flare protocol rewritten as a five-stage return ladder that the phone app now applies automatically to the session it shows you. App gains a full session history with loads, editing, and a progression chart by muscle group.
29 Aug 2026	Pain is now logged as a 0-10 score on both the gym session and the daily check-in, which is what goes to the physio. The traffic light is derived from that number plus the below-the-knee toggle, so the flare ladder and the phase rules are unchanged.
17 October 2026	A side-to-side leg strength difference was found, so every squat, thrust and calf exercise is now single-leg; the hinge (trap bar, back extension, pull-through) stays two-legged on purpose. The leg press is removed entirely because it tensions the sciatic nerve; step-ups move to Day A in Phases 2-3, so single-leg quad work still comes twice a week (step-up on A, Bulgarian split squat on C). Goblet and safety-bar squats are gone, all hip thrusts and the glute bridge are single-leg, calf raises are single-leg, the Phase 2 dumbbell RDL became a B-stance RDL and the Phase 3 walking lunge became the step-up. Added a standing single-leg leg curl (Phases 2-3; there was no knee-flexion hamstring work, and standing keeps the nerve slack) and the cable face pull to Phases 2-3. Curls and pushdowns removed at Andreas's request; dips stay. New benchmark: single-leg strength gap. Added a bodyweight back-extension hold to Day C in Phases 2-3, so the back extensors are trained twice a week: strength on Day B, endurance on Day C. A short-range kneeling ab rollout is added to Phase 3 Day B for heavier anti-extension work; QL stays isometric (side plank, suitcase carry), with no loaded side bends. Phase 3 gets six supersets (marked SS1, SS2…), chosen with Andreas to fit the layout of his gym; the single-leg calf raise moves from Day B to Day A to partner the row.

Sources and further reading

Ameri et al. (2025). *Dynamic Neuromuscular Stabilization Exercise and Chronic Lumbar Disc Herniation: Effects on Pain, Mobility, and Trunk Endurance — A Randomized Controlled Trial.* Health Science Reports. onlinelibrary.wiley.com/doi/10.1002/hsr2.71611

The effect of exercise in the treatment of lumbar disc herniation: a systematic review (2025). PubMed. pubmed.ncbi.nlm.nih.gov/40128486

Effects of Exercise-Based Rehabilitation on Lumbar Degenerative Disc Disease: A Systematic Review (2025). Healthcare, MDPI. mdpi.com/2227-9032/13/15/1938

A phased rehabilitation protocol for athletes with lumbar intervertebral disc herniation. PubMed. pubmed.ncbi.nlm.nih.gov/24175134

Return to advanced strength training and weightlifting in an athlete post-lumbar discectomy utilizing pain neuroscience education and proper progression. IJSPT. pmc.ncbi.nlm.nih.gov/articles/PMC6769273

Basson et al. (2017). *The Effectiveness of Neural Mobilization for Neuromusculoskeletal Conditions: A Systematic Review and Meta-analysis.* JOSPT. jospt.org/doi/10.2519/jospt.2017.7117

Barbell Rehab. *Can You Deadlift With a Herniated Disc?* barbellrehab.com/deadlift-herniated-disc

---

Prepared 31 July 2026, last revised 17 October 2026. This document is educational and does not replace individual clinical assessment. Review it with your physiotherapist before starting, and re-review it after any flare.